PES University – Dept. of CSE

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #02 – Campus & Academic Operations

Automated Rubric Assignment Evaluator

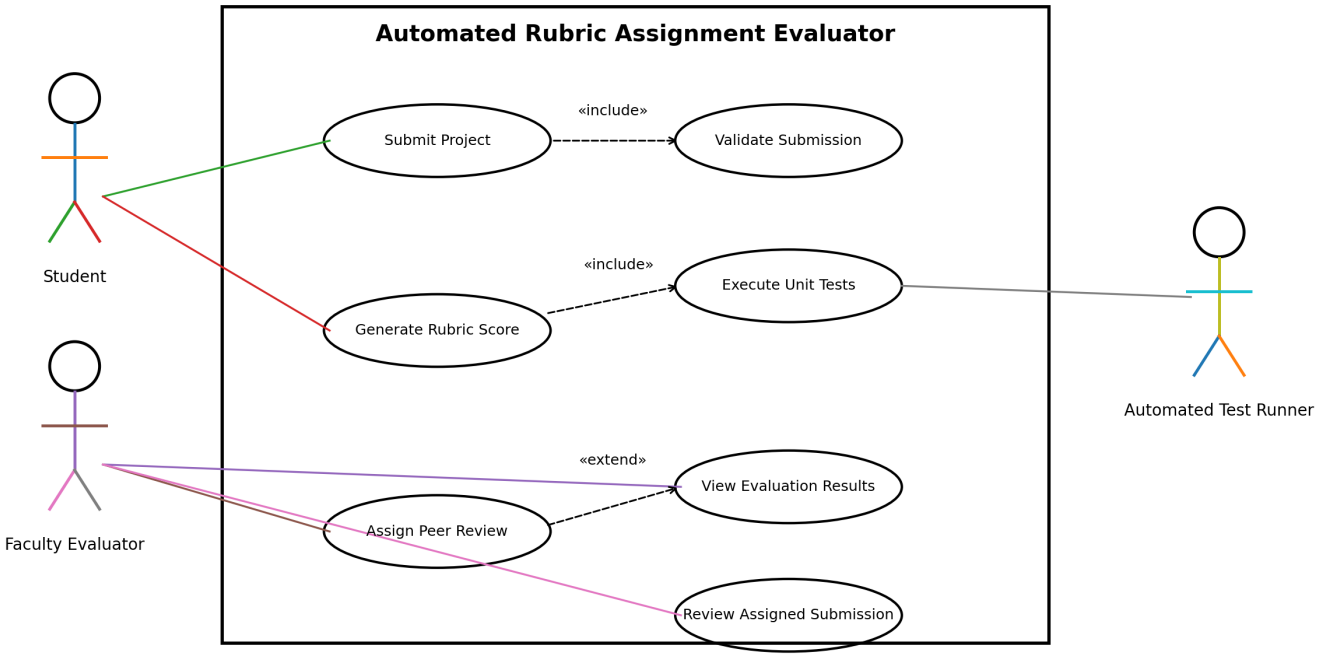
1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall automatically queue uploaded student project ZIP files, execute pre-configured unit tests, and generate an itemized rubric score breakdown.	High	For a valid ZIP submission, the system queues the project, executes all configured tests, and displays a rubric breakdown with test results within 10 seconds of submission.	Core automated evaluation workflow specified in the problem statement.
FR-002	Functional	The system shall validate each uploaded project archive before placing it in the evaluation queue.	High	Given a submission, the system checks file type, archive integrity, required project structure, and size; invalid submissions are rejected with an error message and valid submissions are queued.	Prevents invalid or unusable projects from entering automated evaluation.
FR-003	Functional	The system shall execute the configured test suite against each validated student project and record individual test results.	High	For every validated submission, each configured test is executed and the result is recorded as pass, fail, or error with the corresponding test output.	Provides objective evidence for rubric scoring.
FR-004	Functional	The system shall allow a faculty evaluator to review rubric scores and assign eligible submissions to peer reviewers.	Medium	A faculty evaluator can view a submission's rubric breakdown and assign it to an available peer reviewer; the assignment is stored and visible to the reviewer.	Supports the required peer-review distribution workflow.
FR-005	Functional	The system shall allow students and faculty evaluators to view the evaluation status and final rubric results for a submission.	Medium	After processing, the submission status changes to completed and authorized users can view the test results and itemized rubric score.	Makes evaluation outcomes accessible to the relevant users.
NFR-001	Nonfunctional	The system shall handle up to 100 concurrent project submissions without dropping queue items or exceeding a 1.5 GB memory footprint.	High	A load test with 100 concurrent submissions completes without lost queue items and reports memory usage at or below 1.5 GB.	Ensures reliable performance under the specified simulated load.
NFR-002	Nonfunctional	The system shall protect submitted project files and evaluation results from	High	Security testing confirms that only authorized users can access submission	Protects student code and academic evaluation data.

		unauthorized access during storage and processing.		files/results and that unauthorized access attempts are denied.	

2. UML Use-Case Diagram

Actors: Student, Faculty Evaluator, and Automated Test Runner. The diagram includes the required «include» and «extend» relationships.



Relationship notes:

- Validate Submission is included as part of the submission workflow.
- Execute Unit Tests is included in automated evaluation.
- View Evaluation Results may be extended by peer-review assignment when a review is required.

3. Use-Case Flow Specification

Use Case: Evaluate Student Submission

Primary Actor: Student

Supporting Actors: Automated Test Runner, Faculty Evaluator

Preconditions: Student has a valid project ZIP file; the evaluation service and configured test suite are available; the student is authenticated.

Postconditions: The submission has a recorded evaluation status, test results, and an itemized rubric score. The faculty evaluator can review the result and optionally assign peer review.

Main Success Scenario

1. Student uploads the project ZIP file.

2. System validates the submission archive and required project structure.

3. System places the valid submission in the evaluation queue.

4. System starts the configured unit-test suite for the submission.

5. Automated Test Runner executes the tests and returns individual test results.

6. System maps the test results to the configured rubric criteria.

7. System calculates and stores the itemized rubric score.

8. System marks the submission as evaluated and makes the results available to authorized users.

9. Faculty Evaluator reviews the score and may assign the submission for peer review.

10. Use case ends successfully.

Alternate Flow – Invalid Submission

2a. System detects an invalid, corrupted, or unsupported project ZIP file.

2a1. System rejects the submission and displays the validation error.

2a2. The submission is not added to the evaluation queue.

2a3. Student corrects the project archive and uploads it again.

Source alignment note: The deliverables follow the supplied Lab 1 handout requirement for exactly five FRs and two NFRs, with measurable acceptance criteria, a UML use-case diagram, and a one-page-style use-case flow.